

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

Research Paper / Project Report

Abstract

This project analyzes shipping route efficiency for Nassau Candy Distributor using shipment, customer, product, geographic, sales, cost, and shipping information. The analysis focuses on shipping lead time, route performance, delay frequency, regional bottlenecks, shipment volume, and ship mode performance. Factory locations were assigned to products using the provided product-to-factory mapping, and routes were defined from factory to customer state. A Streamlit dashboard was developed to provide interactive analysis through filters, route comparisons, geographic bottleneck analysis, delay monitoring, and route-level order details. The dataset contains 10,194 shipment records. The calculated average shipping lead time is 1320.84 days, and 33.15% of shipments exceed the selected 1400-day delay threshold. The results provide a structured basis for identifying routes that require monitoring and supporting data-driven logistics planning.

1. Introduction

Efficient transportation is important for controlling delivery delays, managing logistics resources, and maintaining reliable customer service. This project studies factory-to-customer shipping performance for Nassau Candy Distributor. The objective is to identify consistently efficient routes, detect routes with frequent delays, compare shipping performance across regions and shipping modes, and identify geographic areas that may require further investigation.

2. Dataset and Methodology

The dataset contains fields including Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Country/Region, City, State/Province, Postal Code, Division, Region, Product ID, Product Name, Sales, Units, Gross Profit, and Cost. Dates were converted to date format and shipping lead time was calculated as the difference between Ship Date and Order Date. Product names were mapped to the five supplied factory locations. A route was then defined as Factory → Customer State.

For route-level analysis, shipments were grouped by route and summarized using shipment count, average shipping lead time, and lead-time variability. To avoid overinterpreting very small samples, route rankings used a minimum of five shipments where appropriate. Delay frequency was calculated as the percentage of shipments whose lead time exceeded a configurable threshold, with 1400 days used as the default dashboard value.

3. Exploratory Data Analysis and Key Findings

Metric	Result
Total shipments	10,194
Average shipping lead time	1320.84 days
Delay frequency at >1400 days	33.15%
Highest-volume region	Pacific (3,253 shipments)
Highest-volume state	California (2,001 shipments)
Highest average state lead time	West Virginia (1,638 days; 4 shipments)

Highest-volume ship mode	Standard Class (6,120 shipments)
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Regional analysis shows relatively similar average lead times across the four regions, while shipment volume differs. The Pacific region has the highest shipment volume. At state level, California has the largest shipment volume. Some low-volume states show high average lead times, so their results should be interpreted together with shipment counts rather than as standalone evidence of a systematic bottleneck.

Route analysis identified both efficient and slower routes. Among routes with at least five shipments, Secret Factory → Virginia had an average lead time of 1053.4 days, while Lot's O' Nuts → North Dakota had an average lead time of 1638.2 days and a 100% delay frequency under the 1400-day threshold. Lot's O' Nuts → South Dakota and several routes involving New Mexico also showed relatively high delay frequencies. These routes can be prioritized for further operational investigation.

4. Ship Mode Analysis

The dataset contains First Class, Same Day, Second Class, and Standard Class shipping modes. Standard Class has the largest shipment volume at 6,120 shipments and an average lead time of approximately 1314.33 days. First Class has an average of approximately 1338.28 days, Same Day approximately 1333.44 days, and Second Class approximately 1323.85 days. These descriptive results show differences in observed lead times and volumes, but they should not be interpreted as causal evidence that one shipping mode causes faster or slower delivery because route mix and other operational factors may also influence the results.

5. Geographic Bottleneck Analysis

Geographic bottleneck analysis ranks states according to average shipping lead time and provides shipment volume for context. States with high average lead time and sufficient shipment volume can be investigated for possible transportation, scheduling, or network constraints. High-volume states such as California, New York, and Texas are also important from a monitoring perspective because changes in their performance could affect a large number of shipments.

6. Dashboard Development

A Streamlit dashboard was developed using Python, Pandas, and Plotly. The dashboard includes KPI cards for shipment volume, average lead time, and delay frequency. Interactive filters are provided for region, ship mode, order-date range, state, and lead-time threshold. Visual sections include Route Efficiency, Ship Mode Comparison, Regional Shipping Performance, Geographic Bottlenecks, Delay Frequency by Route, Geographic Shipping Map, Route Drill-Down, and Route Performance Details.

7. Recommendations

1. Monitor routes with consistently high average shipping lead times.
2. Investigate routes with high delay frequency, especially when shipment volume is sufficient for reliable comparison.
3. Review high-volume states and regions regularly because they represent a larger share of shipment activity.
4. Compare shipping modes using both lead time and shipment volume before making operational decisions.
5. Focus improvement investigations on routes that combine high shipment volume with high delay frequency.
6. Use the dashboard regularly to monitor geographic bottlenecks and route performance over time.

8. Conclusion

This analysis provides a structured view of factory-to-customer shipping performance for Nassau Candy Distributor. By combining route-level metrics, geographic analysis, delay frequency, and ship mode comparison, the project helps identify areas for operational monitoring and further investigation. The Streamlit dashboard makes these results interactive and allows users to examine performance under different filters and thresholds. Future work could incorporate additional operational variables, validated delivery-time targets, and historical trend analysis to support more detailed logistics optimization.